CHITRANG POTDAR

Backend Engineer | Applied AI

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PROFESSIONAL SUMMARY

Backend Engineer focused on solving the core problem rather than chasing tools. Prioritizes the right architecture and appropriate complexity so systems stay maintainable long after release. Comfortable owning the full lifecycle, from schema to deployment, with a practical focus on AI workflows that complement engineering judgment.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL
Backend: FastAPI, Flask, REST APIs, Celery
Databases & Caching: PostgreSQL, Redis
AI & Agent Frameworks: Claude API, LlamaIndex, LangChain, LangGraph, MCP
Cloud & DevOps: AWS (Lambda, API Gateway, S3, EC2), Docker, GitHub Actions, CI/CD
Third-Party Integrations: SendGrid, Vimeo, Web Push (VAPID)
Observability: Grafana, Loki, Promtail, LangSmith
Testing & Automation: Pytest, Playwright

PROFESSIONAL EXPERIENCE

Backend Engineer · ArrayPointer Pvt. Ltd.

Pune (Oct 2025 – Present)

Investment Operations Platform

Flask · PostgreSQL · Redis · Docker · GitHub Actions · AWS EC2 · Grafana · Loki · Promtail · Pytest · Playwright

- Built a multi-tenant Flask backend with query-level data isolation across users, portfolios, and organizations on a shared PostgreSQL schema; added Redis session caching to reduce redundant DB round-trips, maintaining 80%+ Pytest coverage.
- Developed REST APIs for portfolio review, rejected order tracking, and financial calculators; integrated Jira API to automatically create issues from application events, removing manual reporting.
- Implemented granular RBAC, mapping features to selected domains, with roles assigned to users containing features and their level of access (Create, Read, Update and Delete).
- Automated NSE rejected order processing via Python API integrations and Playwright-driven end-to-end client workflows, saving ~2–3 hrs/week of manual effort.
- Set up the observability stack: Promtail shipping logs to Loki, with Grafana dashboards providing structured visibility into service health and error trends.
- Built a domain-specific MCP server exposing investment platform data to LLM clients, annotating tools/functions with domain terminology so models resolve business concepts without raw schema knowledge.
- Implemented async email delivery via Redis + Celery and VAPID-based Web Push for real-time single-user alerts, including UI failure notifications for unsuccessful email delivery.
- Extended task management with hierarchical ownership/access control across organizational levels; built modules for task reporting, employee performance analytics, and CRM workflows.
- Added authenticated user session tracking and a calendar view covering holidays, meetings, and out-of-office schedules, with task/CRM API restrictions enforcing workflow and availability rules.

LMS & CMS Platform — Jaro Education

FastAPI · PostgreSQL · SendGrid · Vimeo · AWS S3 · Modular Monolithic Architecture

- Designed a modular-monolithic LMS/CMS backend using FastAPI and PostgreSQL, supporting 10K+ students at ~300 RPS, with SendGrid email-verified authentication.
- Built admin-side course creation and student management modules, integrating Vimeo for video upload/streaming.
- Developed student-facing APIs for course access and certificate retrieval, with AWS S3 backing certificate storage.

MahaNRI — NRI Social & Community Platform (Backend Lead)

FastAPI · PostgreSQL · React · AWS · Docker

- Developed backend services for a Maharashtra Government-project social platform targeting the NRI community across multiple applications sharing a common backend.
- Designed and implemented REST APIs across application modules while coordinating API contracts and integration requirements with frontend teams.
- Led backend development activities across a team of developers/interns, including task breakdown, technical guidance, code reviews, PR management, and merge decisions.
- Participated in client discussions to clarify requirements, evaluate technical approaches, and translate product requirements into backend implementation tasks.

- Managed backend deployments and database migrations while coordinating development changes across multiple application modules.

PROJECTS

SOLID-Agent-Harness — Multi-Agent Code Analysis Tool

Node.js · LangChain · LangGraph | github.com/Dev-Chitrang/SOLID-Agent-Harness

- Built a language-agnostic CLI developer-tooling platform running deterministic, graph-based LLM workflows for repository analysis, documentation generation, and code quality — six individual commands (audit, bugs, docs, readme, review, explain) plus five composite workflows.
- Implemented a critic-loop pattern per graph (Agent → Critic → conditional retry) that exits on a quality score threshold or iteration cap, improving output reliability over a single-pass agent call.
- Designed composite commands using LangGraph's parallel fan-out with a barrier-join summary node, running multiple analysis graphs concurrently and synthesizing results into one executive report.
- Built a provider-abstraction layer supporting OpenAI, Anthropic, Gemini, and NVIDIA, swappable without changing any graph code, plus an interactive TUI for provider/command/model selection.
- Architected registry-driven command dispatch so new commands can be added via a folder and registry entry alone, with no changes to core harness or TUI files.

Unified Workspace — Full-Stack Productivity Suite

FastAPI · React · PostgreSQL · Redis · Celery · Docker | github.com/Dev-Chitrang/Productivity-SaaS-Ful-Stack-App

- Built a self-hosted productivity platform unifying task management, notes, calendar scheduling, WebRTC audio meetings, collaborative whiteboarding, and AI meeting analysis into one application.
- Designed a layered Repository → Service → Controller architecture with dependency injection throughout the FastAPI backend for testability and separation of concerns.
- Built a Redis-backed meeting engine with session-scoped lifecycle management and WebSocket signaling for real-time audio meetings.
- Integrated AI-powered meeting analysis via NVIDIA NIM (Llama 3.3 70B) to generate summaries, agenda coverage, and actionable task suggestions from meeting transcripts.
- Implemented JWT authentication with Redis-backed token revocation, OTP-based two-factor auth, and Google OAuth sign-in.
- Maintained 100+ test files with 2,100+ test cases using pytest-asyncio; deployed as a 6-service Dockerized stack.

EDUCATION

B.E. Computer Engineering — 8.65 CGPA

Savitribai Phule Pune University, Pune | 2021 – 2025

RESEARCH & PUBLICATIONS

Corporate Credit Rating Analysis using AI & Sentiment Analysis

IJIRCCE | Impact Factor: 8.65 | Nov 2024